



Introduction to Java Programming

CS 501

Schaefer School of Engineering & Science

Instructor: Debbie Reid

(All assignments are due on Sunday nights at 11:59 p.m. Beijing)

Contact Info: dreid@stevens.edu

Prerequisite(s): None

Corequisite(s): None

Credits: 3

COURSE DESCRIPTION

This course introduces the Java programming language to students with little programming experience. It begins by covering fundamental concepts such as program structure and Java syntax and continues with data types, object-oriented programming, abstract classes and interfaces, control flow, exception handling, recursion, and event-driven programming. Students will write, compile, and execute programs operating on arrays of data or strings, including programs with graphical user interfaces.

STUDENT LEARNING OUTCOMES

After successful completion of this course, you will be able to:

- Apply programming techniques including decision and loop structures, arrays, sorts, string handling, and subprograms to create and execute Java programs.
- Apply object-oriented programming principles including classes, test classes, objects, methods, and exception handling in multiple files.
- Develop graphical user interfaces and components.
- Create a project using advanced JAVA features.

COURSE FORMAT AND STRUCTURE

This course is fully online. To access the course, please visit luminaedu.com/login. For more information about course access or support, contact Beacon staff.

Course Logistics

- You are encouraged to “mentally enroll” in this course as if it occurred on Mondays. In other words, each module will run from Monday to Sunday.
- Written Homework assignments are due every Sunday by 11:59 pm Beijing.
- Multiple choice quizzes will be given at the end of each module.
- Course requirements must be completed and posted or submitted on or before specified due date and delivery time deadline. Due dates and delivery time deadlines are defined as Beijing Time. Please note, students living in distance time zones or overseas must comply with this course time and time and due date deadline policy. Avoid any inclination to procrastinate. To encourage you to stay on schedule, due dates have been established for each assignment; 20% of the total points will be deducted for assignments received at most 2 days late; assignments received more than 2 days late will receive 0 points.
- Each module contains a discussion board. In these boards, you may ask questions about topics pertaining to that module. Do not postcode solutions to the discussion boards unless specifically requested by your instructor.

Instructor's Online Hours

Virtual Office Hours will be held via synchronous Zoom sessions to discuss questions related to weekly readings and/or assignments. The link for the Office Hours Meetings will be posted in the LUMINA Module (Virtual Office Hours). This is OPTIONAL.

TENTATIVE COURSE SCHEDULE

This course is set up based on content acquisition throughout the weeks. Unlike other courses, one module does not equate to one week. During some weeks, you will cover several modules. Please ensure you keep in mind the dates associated with each of the modules to complete the assignments, quizzes, and expectations by the due dates. Assignments posted in one module may be associated with more than one module as well.

Note: All assignments are due on the **last day** for each module. For example, Module 1 and 2 assignments and quizzes are due by 11:59 PM Beijing Time on Sunday.

Week	Module #	Topic(s)/ zyBooks Readings/ Participation Activities	zyBooks Challenge Activities	Programming Assignments Module Quizzes
Week 1	Module 1	Introduction to Java	Challenge Activities 1.2, 1.4	• Introduction • Discussion Post • Module 1 Quiz
	Module 2	Variables / Assignments	Challenge Activities 2.2, 2.5, 2.7, 2.8, 2.9, 2.10, 2.1, 2.12, 2.14, 2.15, 2.18	• Programming Assignment 1 Due • Module 2 Quiz
Week 2	Module 3	Branches	Challenge Activities 3.2, 3.3, 3.4, 3.6, 3.7, 3.10, 3.11, 3.12, 3.13, 3.14, 3.15, 3.16, 3.17	• Programming Assignment 2 Assigned • Module 3 Quiz
	Module 4	Loops	Challenge Activities 4.2, 4.3, 4.4, 4.5, 4.6, 4.8, 4.10, 4.12.	• Programming Assignment 2 Due • Module 4 Quiz
Week 3	Module 5	Arrays	Challenge Activities 5.2, 5.4, 5.5, 5.7.	• Programming Assignment 3 Assigned • Module 5 Quiz
	Module 6	User-Defined Methods	Challenge Activities 6.1,6.2,6.3, 6.4, 6.5, 6.7, 6.8, 6.10.	• Programming Assignment 3 Due • Module 6 Quiz
Week 4	Module 7	Objects and Classes	Challenge Activity 7.3, 7.4, 7.5, 7.8, 7.9, 7.11, 7.14, 7.15, 7.16, 7.19	• Programming Assignment 4 Assigned • Module 7 Quiz
	Module 8	Memory Management	Challenge Activity 8.2	• Programming Assignment 4 Due • Module 8 Quiz

Week 5	Module 9	Input/Output	Challenge Activities 9.2, 9.3	• Programming Assignment 5 Assigned • Module 9 Quiz
	Module 10	Inheritance	Challenge Activities 10.1, 10.3, 10.4, 10.5, 10.6, 10.8, 10.11.	• Programming Assignment 5 Due • Module 10 Quiz
Week 6	Module 11	Exceptions	Challenge Activity 11.1	• Programming Assignment 6 Assigned • Module 11 Quiz
Week 7	Module 12	Searching and Sorting	Challenge Activity 12.6	• Programming Assignment 6 Due • Module 12 Quiz
Week 8	Module 13	GUI/JavaFX Basics	Note: There are no Module 13 Challenge Activities	• Module 13 Quiz

COURSE MATERIALS

zyBook Introduction to Java Programming Online ebook, practice activities, challenge activities) - the link for purchase is provided in the LUMINA course. Additional resources will be provided in LUMINA throughout the term.

COURSE REQUIREMENTS

- **zyBooks Reading/Participation Activities:** Each module contains reading and participation activities from zyBooks. (20%)
- **zyBooks Challenge Activities:** Each module contains challenge activities from zyBooks. (20%)
- **Module Quizzes:** There will be a multiple-choice quiz at the end of each module of the course. (20%)
- **Programming Assignments:** There will be 6 programming assignments that will be submitted outside of zyBooks. (40%)

GUIDELINES FOR DISCUSSIONS

This discussion area will be used as a mechanism to engage with your peers as a post activity to the asynchronous video lectures prior to the live session. This is an active learning opportunity where you will pose and answer questions with regards to the theory of this module.

Please note that there is flexibility in terms of what you post to this area. For example, you can ask questions about topics or concepts where you would like further clarification. You can raise questions about seemingly conflicting issues. Or perhaps you might choose to reflect on and share personal experiences you have had with the concepts or topics covered in this module. Your options are open. The important thing is that you use this area as a mechanism to engage with your peers as you dissect and synthesize the course content.

The **Muddiest Point Discussion Board** is where you can post general questions, discuss areas you are struggling with, or where you just need clarification on a concept that doesn't fit in a specific module. Do not post your code or solutions here, but do ask questions...and if you can answer someone's post please do so. I will monitor this board to be sure the responses are accurate and provide clarification as needed.

TECHNOLOGY REQUIREMENTS

Baseline technical skills necessary for online courses

- Basic computer and web-browsing skills
- Navigating Canvas

Technology skills necessary for this specific course

- Live web conferencing using Canvas Zoom

Required Equipment

- Computer: current Mac (OS X) or PC (Windows 7+) with high-speed internet connection
- Webcam: built-in or external webcam, fully installed
- Microphone: built-in laptop or tablet mic or external microphone

Required Software

- Microsoft Word
- Java SDK
- Eclipse IDE

GRADING PROCEDURES

Grades will be based on:

zyBooks Reading and Participation Activities	20%
zyBooks Challenge Activities	20%
Programming Projects (6)	40%
Module Quizzes (1 per module)	20%

Late Policy

20% of the total points will be deducted for assignments received 1-2 days late; assignments received more than 2 days late will receive 0 points.

There are no makeups for quizzes but your lowest score will be dropped.

Academic Integrity

Graduate Student Code of Academic Integrity

All Stevens graduate students promise to be fully truthful and avoid dishonesty, fraud, misrepresentation, and deceit of any type in relation to their academic work. A student's submission of work for academic credit indicates that the work is the student's own. All outside assistance must be acknowledged. Any student who violates this code or who knowingly assists another student in violating this code shall be subject to discipline.

All graduate students are bound to the Graduate Student Code of Academic Integrity by enrollment in graduate coursework at Stevens. It is the responsibility of each graduate student to understand and adhere to the Graduate Student Code of Academic Integrity. More information including types of violations, the process for handling perceived violations, and types of sanctions can be found at www.stevens.edu/provost/graduate-academics.

Special Provisions for Undergraduate Students in 500-level Courses

The general provisions of the Stevens Honor System do not apply fully to graduate courses, 500 level or otherwise. Any student who wishes to report an undergraduate for a violation in a 500-level course shall submit the report to the Honor Board following the protocol for undergraduate courses, and an investigation will be conducted following the same process for an appeal on false accusation described in Section 8.04 of the Bylaws of the Honor System. Any student who wishes to report a graduate student may submit the report to the Dean of Graduate Academics or to the Honor Board, who will refer the report to the Dean. The Honor Board Chairman will give the Dean of Graduate Academics weekly updates on the progress of any casework relating to 500-level courses. For more information about the scope, penalties, and procedures pertaining to undergraduate students in 500-level courses, see Section 9 of the Bylaws of the Honor System document, located on the Honor Board website.

QUIZ and EXAM CONDITIONS

The following procedures apply to quizzes and exams for this course. As the instructor, I reserve the right to modify any conditions set forth below by printing revised Quiz and Exam Conditions on the quiz or exam.

1. You may use the following materials during quizzes and/or exams. Any materials that are not mentioned in the list below are not permitted.

Material	Permitted?	
	Yes	No
Handwritten Notes	X	
Typed Notes Conditions: i.e. size of note sheet	X	

Textbooks Conditions: i.e. specific books	X	
Readings	X	
Websites (Chegg or similar)		X

2. You are not allowed to work with or talk to other students during quizzes or exams.

Online Etiquette Guidelines

Your instructor and fellow students wish to foster a safe online learning environment. All opinions and experiences, no matter how different or controversial they may be perceived, must be respected in the tolerant spirit of academic discourse. You are encouraged to comment, question, or critique an idea but you are not to attack an individual. Our differences, some of which are outlined in the University's inclusion statement below, will add richness to this learning experience. Please consider that sarcasm and humor can be misconstrued in online interactions and generate unintended disruptions. Working as a community of learners, we can build a polite and respectful course ambience. Please read the Netiquette rules for this course:

Do not dominate any discussion. Give other students the opportunity to join in the discussion.

- Do not use offensive language. Present ideas appropriately.
- Be cautious in using internet language. For example, do not capitalize all letters since this suggests shouting.
- Avoid using vernacular and/or slang language. This could possibly lead to misinterpretation.
- Keep an open mind and be willing to express even your minority opinion.
- Think and edit before you push the "Send" button.
- Do not hesitate to ask for feedback.

LEARNING ACCOMMODATIONS

Stevens Institute of Technology is dedicated to providing appropriate accommodations to students with documented disabilities. The Office of Disability Services (ODS) works with undergraduate and graduate students with learning disabilities, attention deficit-hyperactivity disorders, physical disabilities, sensory impairments, psychiatric disorders, and other such disabilities in order to help students achieve their academic and personal potential. They facilitate equal access to the educational programs and opportunities offered at Stevens and coordinate reasonable accommodations for eligible students. These services are designed to encourage independence and self-advocacy

with support from the ODS staff. The ODS staff will facilitate the provision of accommodations on a case-by-case basis.

For more information about Disability Services and the process to receive accommodations, visit <https://www.stevens.edu/office-disability-services>. If you have any questions please contact: Phillip Gehman, the Director of Disability Services Coordinator at Stevens Institute of Technology at pgehman@stevens.edu or by phone 201-216-3748.

Disability Services Confidentiality Policy

Student Disability Files are kept separate from academic files and are stored in a secure location within the Office of Disability Services. The Family Educational Rights Privacy Act (FERPA, 20 U.S.C. 1232g; 34CFR, Part 99) regulates disclosure of disability documentation and records maintained by Stevens Disability Services. According to this act, prior written consent by the student is required before our Disability Services office may release disability documentation or records to anyone. An exception is made in unusual circumstances, such as the case of health and safety emergencies.

INCLUSIVITY

Name and Pronoun Usage

As this course includes group work and class discussion, it is vitally important for us to create an educational environment of inclusion and mutual respect. This includes the ability for all students to have their chosen gender pronoun(s) and chosen name affirmed. If the class roster does not align with your name and/or pronouns, please inform the instructor of the necessary changes.

Inclusion Statement

Stevens Institute of Technology believes that diversity and inclusiveness are essential to excellence in academic discourse and innovation. In this class, the perspective of people of all races, ethnicities, gender expressions and gender identities, religions, sexual orientations, disabilities, socioeconomic backgrounds, and nationalities will be respected and viewed as a resource and benefit throughout the semester.

Suggestions to further diversify class materials and assignments are encouraged. If any course meetings conflict with your religious events, please do not hesitate to reach out to your instructor to make alternative arrangements.

You are expected to treat your instructor and all other participants in the course with courtesy and respect. Disrespectful conduct and harassing statements will not be tolerated and may result in disciplinary actions.

MENTAL HEALTH RESOURCES

Part of being successful in the classroom involves a focus on your whole self, including your mental health. While you are at Stevens, there are many resources to promote and support mental health. The Office of Counseling and Psychological Services (CAPS) offers free and confidential services to all enrolled students who are struggling to cope with personal issues (e.g., difficulty adjusting to college or trouble managing stress) or psychological difficulties (e.g., anxiety and depression). CAPS is open daily from 9:00 am – 5:00 pm M-F. Evening hours are available by appointment in the Fall / Spring semesters and up-to-date information regarding the availability of evening appointments can be found by visiting www.stevens.edu/CAPS. To schedule an appointment, call 201-216-5177.

Due to the pandemic, in-person appointments may be limited until further notice.

Up-to-date information about the availability of in-person services can be found at www.stevens.edu/CAPS. Teletherapy (therapy via secure video platform) is available to registered students physically located in the states of New York or New Jersey. Students located outside of NY / NJ are encouraged to pursue local treatment through their personal health insurance. To learn more about the process of finding a therapist please visit the CAPS webpage on [Seeking Help Off-Campus](#).

EMERGENCY INFORMATION

In the event of an urgent or emergent concern about the safety of yourself or someone else in the Stevens community, please immediately call the Stevens Campus Police at 201-216-5105 or on their emergency line at 201-216-3911. These phone lines are staffed 24/7, year-round. For students who do not reside near the campus and require emergency support, please contact your local emergency response providers at 911 or via your local police precinct. Other 24/7 national resources for students dealing with mental health crises include the National Suicide Prevention Lifeline (1-800-273-8255) and the Crisis Text Line (text “Home” to 741-741). If you are concerned about the

wellbeing of another Stevens student, and the matter is *not* urgent or time sensitive, please email the CARE Team at care@stevens.edu. A member of the CARE Team will respond to your concern as soon as possible.